Iterative Saliency Aggregation and Assignment Network for Efficient Salient Object Detection in Optical Remote Sensing Images

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***Abstract*— Motivated by the pursuit of efficient salient object detection in remote sensing, researchers have devoted considerable efforts to devising lightweight models due to low running efficiency of the cumbersome models. Although the existing lightweight models have made impressive progress in improving efficiency, there is often a notable sacrifice in inference accuracy, making it challenging to attain both high-quality output and high efficiency. In this article, we propose an iterative saliency aggregation and assignment network (ISAANet) to reconcile the dilemma of balancing accuracy and efficiency. ISAANet adopts a recurrent architecture with bidirectional communication between the encoder and the decoder to boost detection performance. Specifically, a saliency aggregation mechanism is used to generate complementary information by integrating multilevel saliency cues, corresponding to the feedforward information flow from the encoder to the decoder. As for the feedback flows from the decoder to the encoder, a saliency assignment approach is proposed to inject the region and edge prompts into the encoder for hierarchical feature updates, emphasizing the representation of salient information. The encoder is progressively reinforced by alternating iterations of saliency aggregation and assignment, which further helps the decoder to produce more reliable detection results. Moreover, the effective fusion of cross-level features is key to realizing information complementarity and enhancing aggregation quality. However, conventional fusion schemes often neglect feature importance disparities, which can lead to noise accumulation. To address this issue, we introduce a hierarchical interaction and dynamic integration (HIDI) module for feature fusion. The HIDI module groups fusion features and dynamically adjusts their weights based on their information quality, which suppresses noise interference and strengthens the model's learning of meaningful features. The experimental results demonstrate that the proposed ISAANet not only effectively compensates for the accuracy deficiency of the existing lightweight models but also achieves a faster running speed of 108 FPS, which shows a favorable tradeoff between efficiency and accuracy. The code will be available at https://github.com/YiuCK/ISAANet and https://openi.pcl.ac.cn/OpenVision/ISAANet.**

***Index Terms*— Hierarchical interaction and dynamic integration (HIDI), iterative saliency aggregation and assignment, remote sensing, salient object detection.**

I. Introduction

In RECENT years, salient object detection techniques have achieved amazing advancements in natural image research. However, compared with natural images, remote sensing data are characterized by diverse and dense geospatial information, substantial variability in object scales, and unique view angles. Consequently, directly applying mature salient object detection models designed for natural images to remote sensing image analysis may not yield satisfactory performance. Motivated by this challenge, researchers have dedicated their efforts to developing specialized detection techniques tailored for remote sensing scenarios [1], [2], [3], aiming to facilitate intelligent understanding of remote sensing data.

General object detectors such as YOLO [4], [5] are efficient and effective in detecting a wide range of objects and their classes in large images. However, salient object detection has distinct advantages and potential applications that set it apart from traditional object detection. Salient object detection focuses on identifying the most visually striking or important objects in an image, rather than detecting all the objects regardless of their importance. This has significant implications for various applications, as given below.

1) *Efficient Image Compression and Retrieval:* By identifying the most salient objects in an image, compression algorithms can prioritize encoding these regions, resulting in more efficient compression and faster retrieval.
2) *Guidance in Robotics:* In robotic and autonomous systems, saliency detection can direct focus onto the scene's most pertinent elements, fostering more streamlined task execution and informed decision-making.
3) *Adaptive Recognition Through Category-Agnosticism:* A unique strength of salient object detection lies in its category-agnostic nature, enabling it to adaptively

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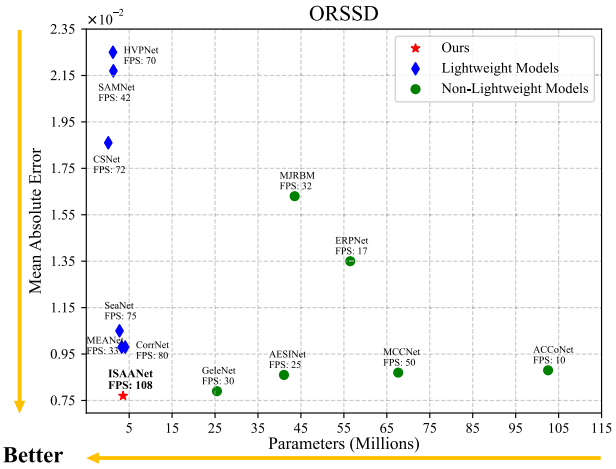


Fig. 1. Comparison of quantitative metrics between lightweight and nonlightweight models on the ORSSD dataset, including model parameters, efficiency, and MAE. Clearly, ISAANet uses a small number of parameters to achieve efficient and high-quality inference.

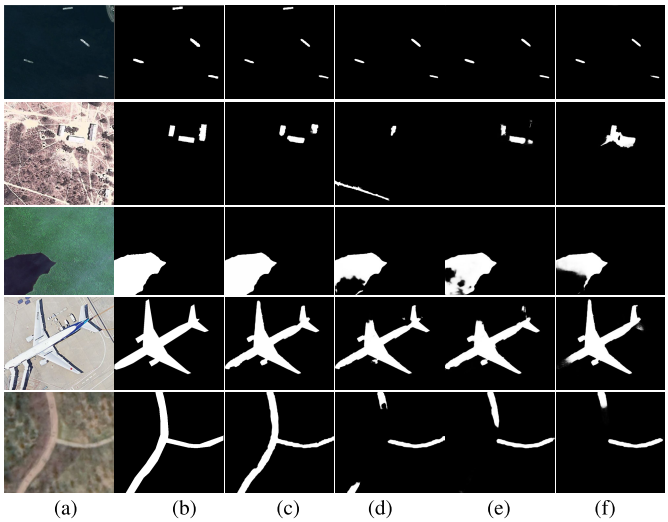


Fig. 2. Saliency maps produced by the existing lightweight models and our model. (a) Input. (b) GT. (c) Ours. (d) MEANet. (e) SeaNet. (f) CorrNet.

recognize visually prominent objects or regions without reliance on predefined class labels. Unlike traditional object detectors, which require training on specific categories and may struggle with novel or unlabeled objects, salient object detection methodologies excel in adapting to new scenarios.

In addition, saliency detection is useful for subjective quality assessment of 3-D graphics [6], [7]. Although object detectors such as YOLO are excellent at detecting multiple objects and their classes, they may not necessarily identify the most visually striking or important objects in an image. By providing a more nuanced understanding of image content, salient object detection enables applications that require a deeper understanding of visual attention and importance, thereby unlocking new possibilities in image analysis and understanding.

In previous research, detection accuracy has been prioritized as the core performance metric, which often leads to a focus on accuracy improvements in model design. To achieve higher accuracy, researchers usually leverage large-scale or complex networks, sometimes without adequately considering the detrimental impacts on inference efficiency. Typically, as model complexity and scale increase, inference speed and efficiency inevitably decrease.

With the rapid development of edge computing technology, lightweight model design has emerged as a prominent and thriving research direction [8], [9], [10]. In the field of remote sensing image processing, lightweight models tailored for salient object detection are burgeoning [11], [12], [13]. The existing models predominantly apply a small-scale backbone to curtail the latency of feature extraction and accelerate inference. However, compared with robust and large-scale backbones [14], [15], [16], [17], lightweight backbones generally exhibit inferior representation capabilities, resulting in a notable accuracy gap between prevailing lightweight models and their cumbersome counterparts, as illustrated in Fig. 1. Besides, as shown in Fig. 2, several typical results of lightweight models exhibit common issues such as incomplete object identification and residual background noise. Therefore, the main challenge currently faced by lightweight models

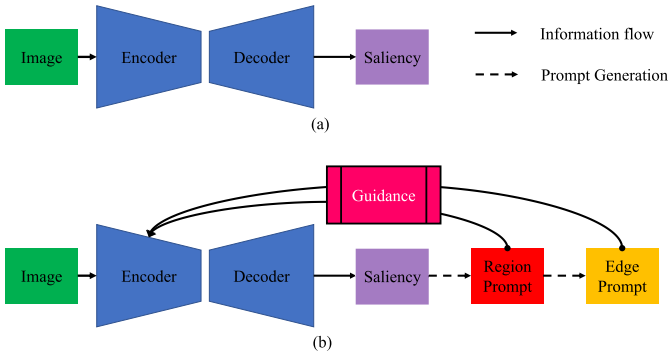


Fig. 3. Difference between the traditional framework of the existing models and the proposed framework. In the traditional framework, information flows unidirectionally from the encoder to the decoder for one-pass inference, with no feedback mechanism. Our proposed framework addresses this limitation by introducing a bidirectional information flow method via cyclic iterations. (a) Unidirectional information flow (traditional framwork). (b) Circular information flow (proposed framwork).

is to find ways to maintain high inference efficiency while minimizing the accuracy gap with cumbersome models.

The existing lightweight models are mainly built upon the conventional inference architecture, as illustrated in Fig. 3(a). The conventional architecture follows the encoder–decoder paradigm, characterized by unidirectional information flow and a single-pass and noniterative inference process. Although the decoder can receive encoded information from the encoder, the encoder lacks real-time feedback from the decoding stage. This absence of bidirectional communication prevents the encoder from dynamically adjusting its output representation of salient features based on the actual requirements of the decoder. Consequently, under such condition, the decoder may be limited by the use of the encoded features that have not been adaptively refined, thus potentially leading to suboptimal inference results.

To address this shortcoming, we design a circular information flow architecture that incorporates a bidirectional communication mechanism, as illustrated in Fig. 3(b). By introducing a feedback pathway from the decoder to the encoder,

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the encoder is allowed to refine its output based on the inference results, thereby strengthening its ability to effectively extract task-relevant features. After undergoing feature updates, the encoder provides more accurate cues to guide the subsequent reasoning process in the decoder, which can gain better accuracy. Specifically, for the remote sensing salient object detection task, considering that multilevel features of the encoder have different attributes, we decompose the feedback from the decoder into region and edge prompts to serve as a more comprehensive guidance mechanism for updating the encoder's representations. Moreover, the synergy between the encoder and the decoder is strengthened by increasing the opportunities for iterative interactions, allowing the saliency information to circulate and be progressively refined within the network. This steadily improves the quality of saliency reasoning over successive cycles.

Specifically, in this article, we propose an iterative saliency aggregation and assignment network (ISAANet). In each iteration, for the ***saliency aggregation*** process, the decoder integrates features from different levels of the encoder to accomplish the segmentation of salient objects. Two types of prompts, representing the salient region and contour knowledge, are then extracted from the current prediction result and assigned to refine the features of corresponding levels in the encoder, which forms the ***saliency assignment*** process. The salient region prompt instructs the deep features to enhance the semantic understanding of specific objects, while the boundary prompt guides the shallow features to focus on the shape details of the targets. After the reinforcement of encoder features, it facilitates the integration of cues for the next round of decoding inference, resulting in more accurate predictions. In addition, simple fusion methods are prone to causing noise superposition among feature layers, which is detrimental to the extraction of complementary information in the saliency aggregation stage. To focus on useful complementary information, we build a hierarchical interaction and dynamic integration (HIDI) module. It adopts a hierarchical fusion mechanism and explores the interaction effects across different features. For each feature interaction, we assess its contribution by calculating its similarity to the set of confidence maps. As a result, combinations with high contribution values will be assigned larger weights. This dynamic adjustment of importance allows the selection of combinations with stronger cooperative effects. Therefore, applying the HIDI modules in the saliency aggregation stage can be effective in fully integrating the various levels of salient cues.

The main contributions of our work can be summarized as follows.

1) We devise a cyclic information flow architecture to facilitate effective bidirectional communication between the encoder and the decoder, which breaks the limitation of unidirectional information flow in the traditional architecture. Based on this framework, we develop an ISAANet, a lightweight model that leverages the prompts from the decoder to refine the encoder in a hierarchical manner. This refinement enables the encoder to concentrate more effectively on salient information. Through multiple rounds of feature reinforcement, the encoder gains the capability to provide more precise saliency cues to the decoder, thereby significantly enhancing inference accuracy.
2) To overcome the defect of traditional cross-layer feature fusion approaches in distinguishing essential complementary information, we introduce a HIDI module. Using a groupwise analysis mechanism, HIDI independently assesses the importance of each group of fusion features and dynamically assigns weights accordingly. This strategic weighting of feature fusion not only diminishes the model's dependency on redundant information but also bolsters its capability to capture critical complementary information, which improves overall performance.
3) Extensive experimental results reveal that our ISAANet model, equipped with the HIDI modules, exhibits outstanding performance on three benchmark datasets. Compared with the recent lightweight models, ISAANet not only achieves a remarkable improvement in detection accuracy, with an average MAE score reduced to 0.0140, but also accelerates the inference speed to 108 FPS. In summary, ISAANet effectively integrates the advantages of both the lightweight and cumbersome models, balancing high efficiency and accuracy to promote its practicality for remote sensing scenarios.

II. Related Work

A. Salient Object Detection for Natural Scene Images

Through the prolonged efforts of researchers, impressive advancements have been achieved in salient object detection for natural scenes [18], [19], [20], [21], [22], [23], [24]. Within the framework of deep learning, scholars commonly use the encoder–decoder structure for image saliency reasoning. Typically, their focus is on the integration of multilevel features [25], [26]. Pang et al. [27] proposed a multiscale interactive network, incorporating aggregate interaction modules to integrate adjacent feature layers and using self-interaction modules for complementary merging of multilevel features. To fuse different features, Li et al. [28] introduced a stacked U-shaped network based on channel attention mechanisms and adopted a multilevel attention cascaded feedback module to focus on salient cues within the feature layers. To enhance the modeling of long-range dependencies, Liu et al. [29] used a Transformer-based backbone for feature extraction, showcasing significant advancements over conventional CNN-based backbones. Moreover, researchers also aim to preserve clear object boundaries in the predictions [30], [31]. To this end, in addition to using salient regions as training labels, boundary labels are incorporated to guide the network in learning boundary information. For example, Wu et al. [32] devised a stacked cross refinement network for the simultaneous detections of salient regions and their boundaries. Qin et al. [33] used a hybrid loss function, comprising BCE, IoU, and SSIM losses, to train the model with the objective of producing high-quality saliency maps.

Current salient object detection models designed for natural scenes predominantly focus on enhancing inference accuracy

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to reliably identify salient objects. To achieve this goal, they often use complex architectures and operations, resulting in low runtime efficiency. Directly applying these models for salient object detection in remote sensing scenes has limited utility. Moreover, given certain intrinsic differences between natural and remote sensing images, the generalization capacities of these models are debatable. When confronted with some complex geographical scenarios, they may struggle to generate accurate saliency maps.

B. Salient Object Detection for Remote Sensing Images

To facilitate the effective application of salient object detection techniques in remote sensing, researchers have begun designing specific models tailored for remote sensing images. Zhang et al. [34] introduced a dense attention fluid network and applied a global context-aware attention module to learn long-range semantic information. Tu et al. [3] developed a multiscale joint region and boundary model to better leverage boundary and region information for reasoning. Similarly, Zhou et al. [35] designed an edge-guided recurrent positioning network to guide the network in learning boundary information to emphasize the maintenance of target boundaries. In addition, to gather precise boundary information to aid in salient region detection, Zeng et al. [36] proposed an adaptive edge-aware semantic interaction network. Furthermore, Li et al. [1] used a multicontent complementation network to acquire complementary clues from both local and global content.

Most of them also adopt the encoder–decoder architecture of unidirectional information flow. Under such a paradigm, the decoding inference relies solely on a one-time encoding of the features. The presence of excessive noise in the encoded features can readily confound model predictions. Moreover, prevailing approaches tend to use large-scale networks in pursuit of high accuracy, compromising efficiency for real-time usage. Our ISAANet establishes a bidirectional interaction between the encoder and the decoder. This allows the decoder to feed back its inference results to the encoder for dynamic adjustment, enabling an innovative iterative optimization process. Besides, in this framework, we use a lightweight encoder to improve inference efficiency.

C. Lightweight Salient Object Detection Models

With the goal to enhance utility for practical usage, an emerging direction is designing lightweight detection models specialized for remote sensing scenes. Li et al. [11] first modified VGG16 [14] to obtain a lightweight backbone, and then adopted a coarse-to-fine strategy to design an inference model called CorrNet. They feed high-level features into a correlation module to produce a coarse saliency map, which is then refined by low-level features to output the final saliency prediction. Compared with previous models, CorrNet reduces model parameters and increases running speed. In addition, Li et al. [12] developed another lightweight model named SeaNet, which uses a semantic matching module and an edge self-alignment module to refine features from different levels, respectively. Although SeaNet further decreases model complexity over CorrNet, no significant accuracy improvement is observed. Liang and Luo [13] also used a lightweight backbone to extract multiscale features, and then constructed an inference model called MEANet. They used an edge embedded attention module, integrating edge information and spatial attention, to enhance representations of salient objects. Overall, MEANet exhibits superior performance over SeaNet. For natural scenes, lightweight detection models have also been proposed, including SAMNet designed by Liu et al. [37] and HVPNet proposed by Liu et al. [38]. Both SAMNet and HVPNet focus on multiscale information learning.

The existing lightweight models are mainly concerned with improving efficiency, but the feature optimization is not well-considered, resulting in obvious performance degradation. They still built upon the conventional encoder–decoder architecture that performs one-time encoding. When the encoded features are polluted by residual noise, decoding accuracy can be hampered. Our approach significantly improves inference precision while maintaining a lightweight design. Through iterative optimization and feature enhancement, our model strikes a superior balance between model compactness and inference accuracy, and the accuracy gap with heavyweight models is further reduced.

D. Review of Advanced Remote Sensing Image Analysis Technologies

Remote sensing technology, as a critical means of acquiring information about the Earth's surface, has a direct impact on the efficiency and accuracy of various domains, including natural resource management and environmental monitoring. In recent years, the rapid development of artificial intelligence (AI) has significantly enhanced the automation and intelligence of remote sensing image analysis. This section aims to provide a comprehensive review of several key remote sensing image analysis tasks, including image classification, hyperspectral anomaly detection, and infrared small target detection, and to explore the latest developments and technical challenges in these fields.

Hong et al. [39] constructed a powerful SpectralGPT framework, designed to unify the processing of multiple downstream tasks, covering scene classification, semantic segmentation, and change detection. The success of SpectralGPT not only pushed the boundaries of AI applications in remote sensing data analysis but also provided a paradigm for building multifunctional, high-efficiency, and large-scale remote sensing processing models. In the field of hyperspectral image analysis, unsupervised anomaly detection [40], [41] technique identifies unique spectral features in an image without a priori information. It has become a key technology in military reconnaissance, precision agriculture, etc. Tu et al. [42] improved the accuracy of anomaly detection by fusing spectral and spatial features of hyperspectral images using a graph model, which highlights the importance of cross-feature dimension fusion. Li et al. [43] combined the low-rank representation model with deep learning technology to further improve the accuracy of anomaly detection. In addition, Li et al. [44] also focused on the hyperspectral imaging task and proposed a CasFormer model to improve the quality of hyperspectral information. Faced with the problems of weak texture and

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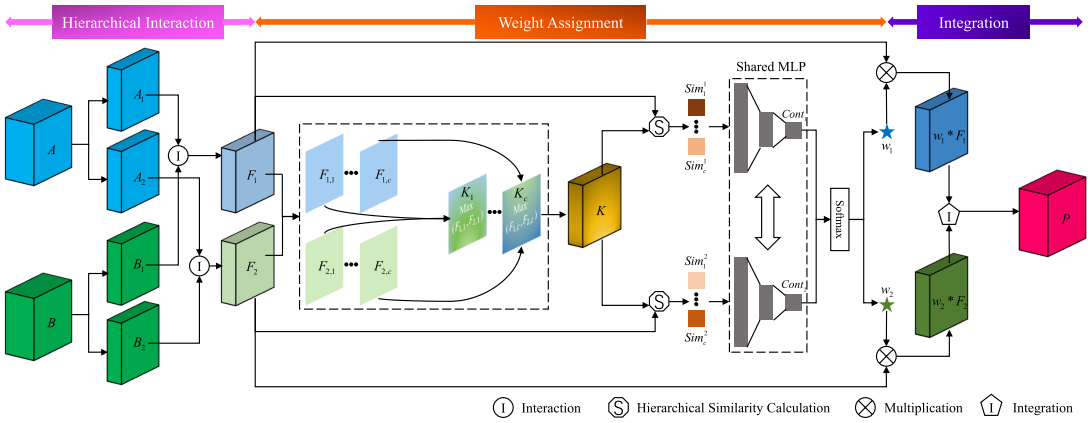


Fig. 4. Structure of the HIDI module. It consists of three stages of information processing, including feature hierarchical interaction, weight assignment, and feature integration.

small size of tiny targets in infrared images, Wu et al. [45] proposed a hybrid UIU-Net model, in which a small UNet is nested within a large UNet to enhance the ability of the model to capture the multilevel features of targets.

Although the various tasks mentioned above face their own unique challenges, the pursuit of efficient feature representation and model generalization capabilities are common goals. Future research should focus on cutting-edge directions such as multitask learning and cross-modal information fusion, to provide more intelligent and precise solutions for solving complex and changing real-world problems.

III. Proposed Method

A. Hierarchical Interaction and Dynamic Integration Module

Feature aggregation techniques are often adopted to consolidate multilevel information for complementary enhancement. However, traditional aggregation fashions, such as element-wise addition and multiplication, encounter two challenges. First, the inability to differentiate useful cues from redundancies results in equal treatment of all fusion outputs, diluting the contribution of critical signals. Second, when vital features are overwhelmed by excessive redundant content, model comprehension of the task could be misled, potentially leading to erroneous inferences. To overcome these limitations, we introduce a hierarchical feature fusion module designed to dynamically distinguish and treat different combinations, as shown in Fig. 4. The core technology is to calculate the differences in salient information in the results of multiple combinations and dynamically assign weights to them based on the quality of the information.

Specifically, given two feature layers $A \in \mathbb{R}^{H\times W\times C}$ and $B \in \mathbb{R}^{H\times W\times C}$ (where C, H, and W represent the channel number, height, and width of the feature maps, respectively), we perform an average channelwise splitting to obtain two sublayer sets, denoted as $A = \{A_1, \ldots, A_N\}$ and $B = \{B_1, \ldots, B_N\}$. Here, N represents the number of sublayers, and each sublayer has the dimensions of $H \times W \times (C/N)$. We adopt a hierarchical fusion strategy across these sublayers to explore varying aggregation effects between A and B, which is formulated as

$$F_i = \text{Conv}_3(\text{Cat}(A_i, B_i)), \quad i = 1, \ldots, N \tag{1}$$

where $F_i \in \mathbb{R}^{H\times W\times (C/N)}$ denotes the fusion result at the ith level, $\text{Cat}(*, *)$ represents the channelwise concatenation, and Conv_3 indicates a 3×3 convolutional layer. To select combinations containing more salient clues from the N combinations and emphasize their contribution, we need to dynamically adjust the importance across combinations. We first compute the salient confidence feature (denoted as $K \in \mathbb{R}^{H\times W\times (C/N)}$) from these combinations, which serves as a criterion to judge aggregation quality. K is calculated as

$$K(u, v, c) = \max\{F_1(u, v, c), \ldots, F_N(u, v, c)\}$$
$$u = 1, \ldots, H; \quad v = 1, \ldots, W; \quad c = 1, \ldots, C/N. \tag{2}$$

Equation (2) calculates the maximum activation value at the same position (u, v) on the same feature channel c. Higher activation reflects stronger model response at that location, thus more likely belonging to regions of interest. The construction of K to some extent considers the distribution patterns of salient region responses, which can be used to quantify the amount of salient information encoded in each combination for quantitative evaluation and selection. Next, we individually compute the correlation of each combination with K to measure its information quality by calculating their channelwise cosine similarity. For the ith aggregation, its similarity (sim_c^i) with K on channel c is calculated as

$$\text{Sim}_c^i = \frac{\sum_{u=1}^{H}\sum_{v=1}^{W}[F_i(u, v, c) \times K(u, v, c)]}{\sqrt{\sum_{u=1}^{H}\sum_{v=1}^{W} F_i(u, v, c)^2} \times \sqrt{\sum_{u=1}^{H}\sum_{v=1}^{W} K(u, v, c)^2}}. \tag{3}$$

Since each interaction F_i has C/N channels, C/N similarity values can be computed between F_i and K. sim_c^i depends on the number of maximum activation values present in the corresponding channel F_c^i. The more maximum values F_c^i contains, the higher sim_c^i is. We feed the set of similarities ($\text{Sim}^i \in \mathbb{R}^{C/N}$) into a multilayer perceptron (MLP) with one hidden layer to obtain the overall contribution score (Cont_i) of F_i toward representing the salient knowledge

$$\text{Cont}_i = \sigma(\text{MLP}(\text{Sim}^i)) = \sigma(W_2\varphi(W_1\text{Sim}^i)) \tag{4}$$

where σ and φ denote the sigmoid and ReLU functions, respectively, and $W_1 \in \mathbb{R}^{(C/N/r)\times(C/N)}$ (r is a reduction factor)

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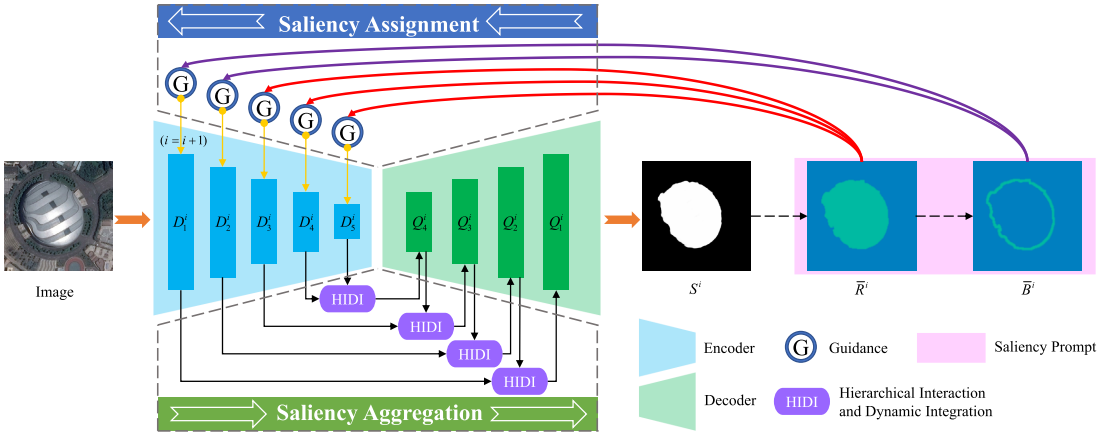


Fig. 5. Multiround iterative reasoning framework of ISAANet. The input image is first encoded by a lightweight backbone to generate multilevel features. Then, these encoded features are aggregated from the deep to shallow direction through multiple HIDI modules. This aggregation process is regarded as performing saliency aggregation. Based on the aggregated result, the saliency map of the first round reasoning can be inferred. Next, salient region and boundary prompts are computed according to the reasoning result from the first round, which are used to update the encoded features of the first round to facilitate the second round reasoning. Specifically, these prompts are used to guide the hierarchical refinement of the encoder, emphasizing relevant semantic cues. This process is viewed as performing saliency assignment. After that, decoding operations for the second round can proceed to produce the saliency map. Overall, the ISAANet can achieve high-quality saliency reasoning by iteratively performing saliency aggregation and assignment across multiple rounds.

and $W_2 \in \mathbb{R}^{1\times(C/N/r)}$. A larger Cont_i value indicates that F_i contains richer salient information, warranting a higher fusion weight to have a more significant impact. After these steps, Cont_i for each interaction is dynamically determined. We normalize the contributions across all combinations to obtain their weights (w_i) for final fusion

$$w_i = \frac{\text{Cont}_i}{\sum_{i=1}^{N} \text{Cont}_i}. \tag{5}$$

According to (5), the weight of a combination is positively correlated with its contribution value. This achieves dynamic weight assignment based on the quantitative disparities in combination efficacy. After obtaining the interaction set $\{F_1, F_2, \ldots, F_N\}$ and the associated weight set $\{w_1, w_2, \ldots, w_N\}$, we consolidate these two sets to reorganize the internal resources within the interaction set to generate fusion result $P \in \mathbb{R}^{H\times W\times C}$

$$P = \text{Conv}_3(\text{Conv}_1(\text{Cat}(w_1 \times F_1, \ldots, w_N \times F_N))) \tag{6}$$

where Conv_1 represents a 1×1 convolutional layer. Overall, as shown in Fig. 4, the information processing in the HIDI module consists of three steps, including feature interaction, weight assignment, and feature integration. The final resulting feature layer P is used as the output of HIDI module.

B. Iterative Saliency Aggregation and Assignment Network

In salient object detection models for remote sensing images, the conventional encoder–decoder structure suffers from unidirectional information flow, easily causing inferior decoding due to one-time encoding operations. Particularly when the foreground–background distinction is insufficient in the encoded features, recognition performance can be interfered by noise. To address this issue, we propose an ISAANet via establishing bidirectional communication between the encoder and the decoder. By gradually optimizing the encoded features to accentuate salient knowledge, the model enriches salient region expression iteratively. With the bidirectional communication between the encoder and the decoder, decoding capability is continuously improved to effectively tackle challenges in detecting salient objects from remote sensing images.

The architecture of ISAANet is illustrated in Fig. 5. Formally, an input image $I \in \mathbb{R}^{H_I\times W_I\times 3}$ is processed by the lightweight MobileNetV3 [46], and five feature layers at different scales are produced. The resolutions of these feature layers are $(H_I/2^i) \times (W_I/2^i), i = 1, 2, 3, 4, 5$, with channel numbers sequentially set to 16, 24, 40, 112, and 960. For subsequent fusion convenience, we unify their channel numbers to 32, thereby obtaining the initial encoded feature set $D^1 = \{D_1^1, D_2^1, D_3^1, D_4^1, D_5^1\}$. This initial multilevel feature extraction provides a rich semantic basis to facilitate the iterative optimization process.

To implement the first-stage saliency inference, we perform progressive decoding and clue extraction from the encoded feature set D^1. To this end, multilevel features in D^1 are fused from the deep to shallow direction. We use the designed HIDI module for interlayer feature fusion

$$Q_i^1 = \begin{cases} \text{HIDI}\left(D_i^1, \uparrow\left(D_{i+1}^1\right)\right), & i = 4 \\ \text{HIDI}\left(D_i^1, \uparrow\left(Q_{i+1}^1\right)\right), & i = 1, 2, 3 \end{cases} \tag{7}$$

where $\uparrow(*)$ denotes feature map upsampling. It is important that the fusion result Q_4^1 of D_4^1 and D_5^1 is calculated first. Subsequently, the lower level features are progressively fused onto this result. This step achieves the aggregation of multilevel salient clues, termed as saliency aggregation. Following this, the first-stage saliency map S^1 can be derived from Q_1^1:

$$S^1 = \sigma\left(\text{Conv}_1\left(\text{Conv}_3\left(Q_1^1\right)\right)\right) \tag{8}$$

where 1×1 and 3×3 convolutional layers output 1 and 32 channels, respectively. Next, to optimize the encoding input

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for the second stage, we use S^1 as the feedback to enhance the representations of features in Q_1^1, accentuating saliency-related information. This process can be viewed as introducing prior knowledge to guide encoder adjustment.

We first process the first-stage prediction S^1 to derive salient region prompt (R^1) and boundary prompt (B^1). Specifically, S^1 is dynamically binarized to generate foreground region prompt R^1

$$R^1(u,v) = \begin{cases} 1, & S^1(u,v) \geq T \\ 0, & S^1(u,v) < T \end{cases} \tag{9}$$

where T is a threshold. The meaning of (9) is that if the foreground probability at a spatial location in S^1 is greater than T, it is regarded as a salient pixel, and vice versa as a background pixel. In R^1, the foreground regions are segmented, and we can use R^1 to enhance the deep features of D^1 so that they can highlight semantic information of salient objects. Moreover, to make the shallow features of D^1 focus on the detailed structures of the targets, we use the boundary prompt to refine them. The steps of generating B^1 are computed as follows:

$$\bar{R}^1 = \psi(R^1) \tag{10}$$

$$B^1 = \bar{R}^1 \ominus R^1 \tag{11}$$

where $\psi(*)$ denotes the dilation operation with a 3×3 kernel size, and $\ominus$ represents elementwise subtraction. In practice, we do not directly use R^1 and B^1 to refine deep and shallow features, respectively. Instead, we perform dilation on them to obtain smoothly expanded regions $\bar{R}^1$ and $\bar{B}^1(\bar{B}^1 = \psi(B^1))$. This practice offers dual benefits. On one hand, dilation moderately enlarges the spatial coverage of the target regions, facilitating object localization for the network. On the other hand, the expanded boundary details provide the network with smoother transitions from the foreground to background, simultaneously enhancing awareness of target boundaries.

We categorize the encoded features in D^1 into shallow features (D_1^1 and D_2^1) and deep features (D_3^1, D_4^1, and D_5^1). Considering they represent detail and semantic information, respectively, we use salient prompts to selectively update them for enhancing associated cues. This process, termed as saliency assignment, produces the encoded feature set for the second stage, denoted as

$$D_i^2 = \begin{cases} \text{Conv}_3\left(\text{Conv}_1\left(D_i^2 \oplus D_i^2 \otimes \overline{B}^1\right)\right), & i = 1,2 \\ \text{Conv}_3\left(\text{Conv}_1\left(D_i^2 \oplus D_i^2 \otimes \overline{R}^1\right)\right), & i = 3,4,5 \end{cases} \tag{12}$$

where $\oplus$ and $\otimes$ represent elementwise addition and multiplication, respectively. Fig. 6 illustrates examples of feature enhancement effects. It can be observed that with the guidance of the corresponding prior knowledge, encoded features at different levels can more effectively emphasize foreground regions or boundary structures, providing richer multigranularity salient cues for subsequent decoding. After obtaining the encoded feature set for the second stage, decoding fusion and saliency prediction are conducted similarly as follows:

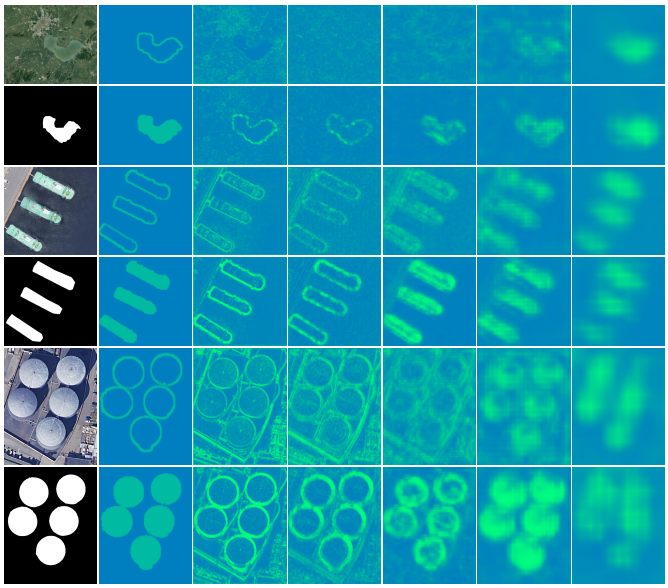


Fig. 6. Visual comparisons of the original features and the enhanced features. Here, I, GT, EP, and RP represent the input image, ground truth, edge prompt, and region prompt, respectively. (a) I and GT. (b) EP and RP. (c) D_1^1 and D_1^2. (d) D_2^1 and D_2^2. (e) D_3^1 and D_3^2. (f) D_4^1 and D_4^2. (g) D_5^1 and D_5^2.

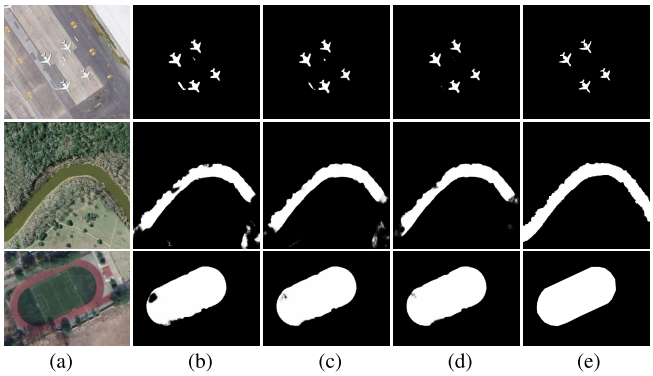


Fig. 7. Saliency maps generated at various stages of ISAANet. Here, we set $Z = 3$, and therefore, ISAANet outputs three saliency maps during inference. (a) Input. (b) S^1. (c) S^2. (d) S^3. (e) GT.

$$Q_i^2 = \begin{cases} \text{HIDI}\left(D_i^2, \uparrow \left(D_{i+1}^2\right)\right), & i = 4 \\ \text{HIDI}\left(D_i^2, \uparrow \left(Q_{i+1}^2\right)\right), & i = 1,2,3 \end{cases} \tag{13}$$

$$S^2 = \sigma\left(\text{Conv}_1\left(\text{Conv}_3\left(Q_1^2\right)\right)\right). \tag{14}$$

By iterating the above saliency aggregation and assignment steps, we can construct an iterative framework. Each iteration (except for the first one) uses the feedback from the previous round of prediction to adjust the expression of the encoded features, gradually refining the effect of salient object detection. The prediction results at different numbers of iterations rounds (defined as Z) are presented in Fig. 7. It is evident that as the number of iterations increases, the salient regions in the prediction become more coherent and uniform, concurrently suppressing noise in the background regions. This validates that salient information gets extracted and refined in a gradual

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process within the network. Ultimately, we select the output from the last iteration as the result of ISAANet.

C. Loss Function

ISAANet contains Z stages of inference, correspondingly outputting Z saliency maps. Therefore, we use ground truth to supervise the predictions at each stage. The loss function for a saliency map consists of binary cross-entropy (BCE) and intersection over union (IoU) losses, and they are defined as follows:

$$\begin{aligned} &L_{\mathrm{BCE}}(S^i, G) \\ &= -\sum_{u,v} G(u,v)\log S^i(u,v) \\ &\quad + [1 - G(u,v)]\log[1 - S^i(u,v)] \end{aligned} \tag{15}$$

$$\begin{aligned} &L_{\mathrm{IoU}}(S^i, G) \\ &= 1 - \frac{\sum_{u=1}^{H_I}\sum_{v=1}^{W_I} S^i(u,v)\cdot G(u,v)}{\sum_{u=1}^{H_I}\sum_{v=1}^{W_I}[S^i(u,v) + G(u,v) - S^i(u,v)\cdot G(u,v)]} \end{aligned} \tag{16}$$

where S^i is the saliency map of the ith stage, and G is the label of the salient region. The overall loss of the network is denoted as

$$L^{\mathrm{total}} = \sum_{i=1}^{Z} L_{\mathrm{BCE}}(S^i, G) + \lambda L_{\mathrm{IoU}}(S^i, G) \tag{17}$$

where λ is used to balance the effects of the two losses.

IV. Experiments

A. Experimental Setup

1) Dataset: To evaluate the model performance, experiments are conducted on three public remote sensing datasets: ORSSD [47], EORSSD [34], and ORSI-4199 [3]. They contain both training and test sets with annotations for network supervision and performance evaluation. The ORSSD dataset is the first dataset tailored for remote sensing salient object detection, with 600 images in the training set and 200 images for the test set. The EORSSD dataset expands ORSSD by adding more samples, resulting in 1400 and 600 images for training and test sets, respectively. The ORSI-4199 dataset is the largest among them, with a total of 4199 images. Specifically, 2000 images are used for training, while the remaining 2199 images constitute the test set. These datasets showcase a variety of object scales, presenting challenges for salient object detection, especially in images with complex backgrounds or multiple salient objects. By convention, we use the training set in each dataset for model training and evaluate performance on the respective test sets.

2) Implementation Detail: We build ISAANet with PyTorch and apply a NVIDIA GPU (RTX 3090) to accelerate inference. For training, we use random rotation and horizontal flipping for data augmentation. The batch size is set to 16, with network parameters updated using the Adam optimizer. The initial learning rate is 0.0004, gradually reduced following a cosine annealing schedule. The training ran for 85 epochs. In the HIDI module, we set $N = 2$ and $r = 4$. Moreover, the inference iteration number Z of ISAANet is set to 3. In (9), the threshold T is set to 0.2. As for the balancing factor λ in (17), we set it to 3. During both training and test, the input image resolution is resized to 352 × 352 pixels.

3) Evaluation Metrics: Given the well-established evaluation metrics for natural image salient object detection models, we follow the existing work and use these metrics to assess model performance in the remote sensing domain. These metrics gauge the quality of saliency maps, including F-measure (F_β) [48], S-measure (S_α) [49], E-measure (E_ξ) [50], and mean absolute error (MAE, $\mathcal{M}$). For F-measure and E-measure, we record their maximum, mean, and adaptive scores. Higher scores, except for MAE, signify superior model performance. Besides, we use more intuitive metrics including precision ($\mathcal{P}$), recall ($\mathcal{R}$), and (IoU) to further reveal the detection performance of the models. In addition to saliency map evaluations, we also consider model size and runtime efficiency as key metrics determining the model's deployment effect. Consequently, we report pertinent metrics such as parameter counts, floating-point operations (FLOPs), and frames per second (FPS). A comprehensive evaluation of a model's inference quality and practicality is achieved by considering the aforementioned metrics.

B. Comparison With State-of-the-Arts

To highlight the superior performance of ISAANet, we compare it with several state-of-the-art models in the field of remote sensing salient object detection, including MCCNet [1], MJRBM [3], ERPNet [35], ACCoNet [51], AESINet [36], and GeleNet [52]. In addition, we choose famous detection models designed for natural scenes, such as SUCA [28], PA-KRN [53], VST [29], DNTDF [54], and ICON [55], which have been fine-tuned on remote sensing data for evaluation against ISAANet. Furthermore, we select lightweight detection models such as CSNet [56], SAM-Net [37], HVPNet [38], CorrNet [11], SeaNet [12], and MEANet [13] to investigate model complexity and speed metrics. In addition to the saliency detection models, we also introduce infrared small target detection models for model comparison, including UIU-Net [45] and MSHNet [57]. The prediction results for all the compared models are either directly acquired from the authors or generated by running available code. To ensure the fairness and comparability of experiments, we strictly adhered to uniform experimental standards. Specifically, the compared models were retrained on the same training dataset, and consistent random seed values were used during the training process to ensure consistency in dataset splitting, thereby eliminating potential biases introduced by data shuffling differences.

1) Quantitative Comparison: Fig. 8 shows the F-measure curves for all the models on the three datasets. Our ISAANet curve (marked in red) lies above most curves of existing models and consistently achieves high F-measure scores across most thresholds. Thus, ISAANet has advanced detection capabilities. In addition, Tables I and II further provide a detailed performance comparison of various models on three datasets.

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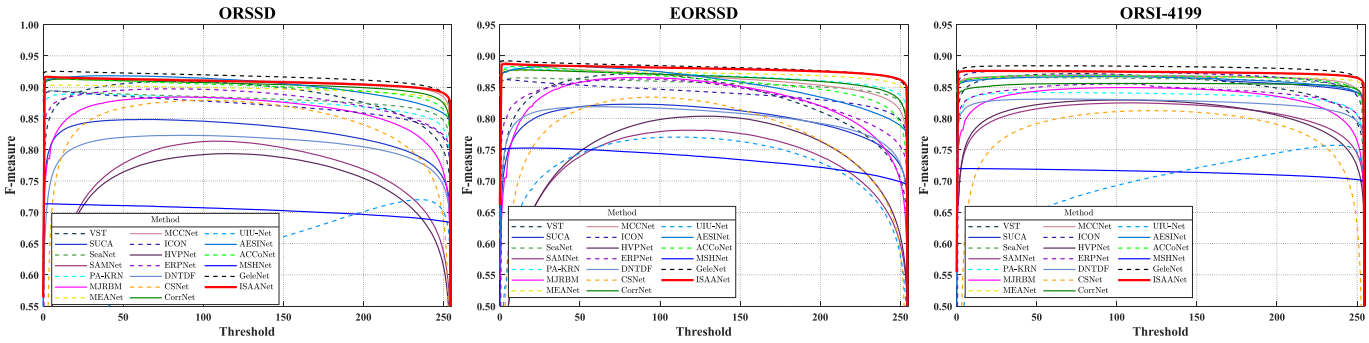


Fig. 8. F-measure curves of the existing methods and our methods on the three datasets.

TABLE I

PERFORMANCE EVALUATION OF 18 MODELS ON THE ORSSD AND EORSSD DATASETS. FOR LIGHTWEIGHT MODELS, THE TOP THREE SCORES ARE MARKED IN **RED**, **GREEN**, AND **BLUE**, RESPECTIVELY

Methods	Type	Publication Year	Speed (FPS)↑	#Param (M)↓	FLOPs (G)↓	ORSSD [47]											EORSSD [34]										
						$S_\alpha\uparrow$	$E_\xi^{adp}\uparrow$	$E_\xi^{mean}\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{adp}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$\mathcal{P}\uparrow$	$\mathcal{R}\uparrow$	$IoU\uparrow$	$S_\alpha\uparrow$	$E_\xi^{adp}\uparrow$	$E_\xi^{mean}\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{adp}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$\mathcal{P}\uparrow$	$\mathcal{R}\uparrow$	$IoU\uparrow$
SUCA [28]	C.N	TMM 21	24	117.71	56.4	0.8989	0.9194	0.9400	0.9584	0.7748	0.8237	0.8484	0.0145	0.7682	0.9520	0.7598	0.8988	0.9082	0.9277	0.9520	0.7260	0.7949	0.8229	0.0097	0.7222	0.9450	0.7194
PA-KRN [53]	C.N	AAAI 21	16	141.06	617.7	0.9239	0.9579	0.9620	0.9680	0.8548	0.8727	0.8890	0.0139	0.8608	0.9441	0.8237	0.9293	0.9680	0.9675	0.9787	0.8406	0.8630	0.8812	0.0111	0.8524	0.9544	0.8276
VST [29]	T.N	ICCV 21	23	44.09	23.2	0.9365	0.9466	0.9621	0.9810	0.8262	0.8817	0.9095	0.0094	0.8200	**0.9743**	0.8244	0.9208	0.8941	0.9442	0.9743	0.7089	0.8263	0.8716	0.0067	0.6972	**0.9728**	0.7659
DNTDF [54]	C.N	SCIS 22	31	28.84	10.9	0.8698	0.9081	0.9086	0.9286	0.7645	0.8020	0.8231	0.0217	0.7745	0.9069	0.7116	0.8957	0.9047	0.9225	0.9378	0.7288	0.7962	0.8189	0.0113	0.7306	0.9377	0.7320
ICON [55]	T.N	TPAMI 23	25	33.1	20.97	0.9256	0.9554	0.9637	0.9704	0.8444	0.8671	0.8939	0.0116	0.8445	0.9528	0.8268	0.9185	0.9497	0.9619	0.9687	0.8065	0.8371	0.8622	0.0073	0.8121	0.9526	0.7998
MCCNet [1]	C.R	TGRS 22	50	67.65	112.8	0.9437	0.9735	0.9758	0.9800	0.8957	0.9054	0.9155	0.0087	0.9108	0.9440	0.8663	0.9327	0.9538	0.9685	0.9755	0.8137	0.8604	0.8904	0.0066	0.8204	0.9570	0.8350
MJRBM [3]	C.R	TGRS 22	32	43.54	95.7	0.9204	0.9328	0.9415	0.9623	0.8022	0.8566	0.8842	0.0163	0.7989	0.9503	0.8185	0.9197	0.8897	0.9350	0.9646	0.7066	0.8239	0.8656	0.0099	0.6948	0.9655	0.7632
ERPNet [35]	C.R	TCYB 23	17	56.48	215.4	0.9254	0.9520	0.9566	0.9710	0.8356	0.8745	0.8974	0.0135	0.8424	0.9312	0.8526	0.9210	0.9228	0.9401	0.9603	0.7554	0.8304	0.8632	0.0089	0.7542	0.9445	0.7710
ACCoNet [51]	C.R	TCYB 23	10	102.55	348.8	0.9437	0.9721	0.9754	0.9796	0.8806	0.8971	0.9149	0.0088	0.8871	0.9555	0.8637	0.9290	0.9450	0.9653	0.9727	0.7969	0.8552	0.8837	0.0074	0.7972	0.9602	0.8230
AESINet [36]	C.R	TGRS 23	25	41.05	55.0	0.9460	0.9707	0.9747	0.9828	0.8666	0.8986	0.9183	0.0086	0.8693	0.9586	0.8578	0.9358	0.9462	0.9636	0.9751	0.7923	0.8524	0.8838	0.0079	0.7952	0.9586	0.8122
UIU-Net [52]	C.R	TIP 23	35	50.54	85.16	0.7643	0.8770	0.7444	0.8913	0.6978	0.6244	0.7202	0.0341	0.7214	0.7895	0.5134	0.8480	0.8020	0.8612	0.9119	0.5899	0.7198	0.7702	0.0189	0.5813	0.9202	0.6253
GeleNet [52]	T.R	TIP 23	30	25.45	11.7	**0.9469**	**0.9786**	**0.9815**	**0.9860**	**0.9035**	**0.9128**	**0.9254**	0.0079	**0.9161**	0.9553	**0.8797**	**0.9376**	**0.9750**	**0.9766**	**0.9828**	0.8641	**0.8781**	**0.8923**	0.0064	0.8813	0.9514	**0.8487**
CSNet [56]	L.N	ECCV 20	72	0.14	0.7	0.8910	0.9068	0.9171	0.9628	0.7615	0.8285	0.8790	0.0186	0.7529	0.9627	0.7542	0.8364	0.8339	0.8929	0.9535	0.6319	0.7656	0.8341	0.0169	0.6181	0.9591	0.6526
SAMNet [37]	L.N	TIP 21	42	1.33	0.5	0.8761	0.8656	0.8818	0.9478	0.6843	0.7531	0.8137	0.0217	0.6684	0.9471	0.6568	0.8622	0.8284	0.8700	0.9421	0.6114	0.7214	0.7813	0.0132	0.5930	0.9501	0.6199
HVPNet [38]	L.N	TCYB 21	70	1.23	1.1	0.8610	0.8471	0.8717	0.9320	0.6726	0.7396	0.7938	0.0225	0.6526	0.9547	0.6466	0.8734	0.8270	0.8721	0.9482	0.6202	0.7377	0.8036	0.0110	0.6013	0.9570	0.6323
CorrNet [11]	L.R	TGRS 22	80	4.09	21.1	0.9380	0.9721	0.9746	0.9790	0.8875	0.9002	0.9129	0.0098	0.9000	0.9411	0.8565	0.9289	0.9593	0.9646	0.9696	0.8311	0.8620	0.8778	0.0083	0.8481	0.9370	0.8219
SeaNet [12]	L.R	TGRS 23	75	2.76	1.7	0.9260	0.9670	0.9722	0.9767	0.8625	0.8772	0.8942	0.0105	0.8709	0.9314	0.8270	0.9208	0.9602	0.9651	0.9710	0.8304	0.8519	0.8649	0.0073	0.8491	0.9218	0.7978
MEANet [13]	L.R	ESWA 24	33	3.27	9.62	0.9340	0.9715	0.9730	0.9768	0.8860	0.8934	0.9033	0.0098	0.9029	0.9355	0.8492	0.9282	0.9663	0.9658	0.9708	0.8557	0.8678	0.8766	0.0070	0.8824	0.9228	0.8222
MSHNet [52]	L.R	CVPR 24	85	4.07	6.11	0.8128	0.8694	0.8782	0.8831	0.6895	0.7010	0.7136	0.0519	0.6949	0.8298	0.7036	0.8543	0.8866	0.8962	0.9024	0.7000	0.7325	0.7528	0.0251	0.7126	0.8772	0.6727
ISAANet (Ours)	L.R	TGRS 24	108	3.59	4.34	**0.9429**	**0.9777**	**0.9787**	**0.9836**	**0.8982**	**0.9034**	**0.9166**	**0.0077**	**0.9113**	0.9516	**0.8692**	**0.9351**	**0.9734**	**0.9717**	**0.9761**	**0.8643**	**0.8748**	**0.8869**	**0.0063**	**0.8872**	0.9377	**0.8381**

C.N: CNN-based models for natural scenes, T.N: Transformer-based models for natural scenes, C.R: CNN-based models for remote sensing,
T.R: Transformer-based models for remote sensing, L.N: lightweight models for natural scenes, L.R: lightweight models for remote sensing.

TABLE II

PERFORMANCE EVALUATION OF 18 MODELS ON THE ORSI-4199 DATASET. FOR LIGHTWEIGHT MODELS, THE TOP THREE SCORES ARE MARKED IN **RED**, **GREEN**, AND **BLUE**, RESPECTIVELY

Methods	Type	Publication Year	Speed (FPS)↑	#Param (M)↓	FLOPs (G)↓	ORSI-4199 [3]										
						$S_\alpha\uparrow$	$E_\xi^{adp}\uparrow$	$E_\xi^{mean}\uparrow$	$E_\xi^{max}\uparrow$	$F_\beta^{adp}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{max}\uparrow$	$\mathcal{M}\downarrow$	$\mathcal{P}\uparrow$	$\mathcal{R}\uparrow$	$IoU\uparrow$
SUCA [28]	C.N	TMM 21	24	117.71	56.4	0.8794	0.9186	0.9356	0.9438	0.8415	0.8590	0.8692	0.0304	0.8560	0.8605	0.7645
PA-KRN [53]	C.N	AAAI 21	16	141.06	617.7	0.8491	0.9063	0.9168	0.9280	0.8200	0.8324	0.8415	0.0382	0.8403	0.8242	0.7277
VST [29]	T.N	ICCV 21	23	44.09	23.2	0.8790	0.8997	0.9348	0.9481	0.7947	0.8524	0.8717	0.0281	0.8142	**0.8820**	0.7602
DNTDF [54]	C.N	SCIS 22	31	28.84	10.9	0.8444	0.8963	0.9050	0.9158	0.8065	0.8208	0.8310	0.0425	0.8234	0.8366	0.7071
ICON [55]	T.N	TPAMI 23	25	33.1	20.97	0.8752	0.9239	0.9438	0.9521	0.8531	0.8664	0.8763	0.0282	0.8615	0.8656	0.7698
MCCNet [1]	C.R	TGRS 22	50	67.65	112.8	0.8746	0.9182	0.9348	0.9413	0.8592	0.8630	0.8690	0.0316	0.8730	0.8580	0.7699
MJRBM [3]	C.R	TGRS 22	32	43.54	95.7	0.8593	0.8891	0.9102	0.9311	0.7995	0.8309	0.8493	0.0374	0.7988	0.8748	0.7246
ERPNet [35]	C.R	TCYB 23	17	56.48	215.4	0.8670	0.9024	0.9149	0.9290	0.8024	0.8374	0.8553	0.0357	0.8294	0.8444	0.7357
ACCoNet [51]	C.R	TCYB 23	10	102.55	348.8	0.8775	0.9167	0.9342	0.9412	0.8581	0.8620	0.8686	0.0314	0.8683	0.8688	0.7704
AESINet [36]	C.R	TGRS 23	25	41.05	55.0	0.8702	0.9184	0.9310	0.9403	0.8251	0.8573	0.8670	0.0325	0.8646	0.8312	0.7567
UIU-Net [52]	C.R	TIP 23	35	50.54	85.16	0.7509	0.8516	0.7706	0.8795	0.7296	0.6887	0.7571	0.0688	0.7815	0.7163	0.5368
GeleNet [52]	T.R	TIP 23	30	25.45	11.7	**0.8862**	**0.9265**	**0.9478**	**0.9544**	**0.8755**	**0.8785**	**0.8842**	**0.0264**	**0.8894**	0.8656	**0.7871**
CSNet [56]	L.N	ECCV 20	72	0.14	0.7	0.8241	0.8447	0.8586	0.9096	0.7162	0.7674	0.8124	0.0524	0.7330	0.8456	0.6480
SAMNet [37]	L.N	TIP 21	42	1.33	0.5	0.8409	0.8781	0.8938	0.9186	0.7744	0.8029	0.8249	0.0432	0.7758	0.8557	0.6838
HVPNet [38]	L.N	TCYB 21	70	1.23	1.1	0.8471	0.8687	0.8956	0.9201	0.7652	0.8041	0.8295	0.0419	0.7602	0.8742	0.6931
CorrNet [11]	L.R	TGRS 22	80	4.09	21.1	0.8623	0.9142	0.9206	0.9330	0.8534	0.8513	0.8560	0.0366	0.8805	0.8269	0.7448
SeaNet [12]	L.R	TGRS 23	75	2.76	1.7	0.8722	0.9197	0.9363	0.9426	0.8556	0.8591	0.8653	0.0308	0.8707	0.8550	0.7632
MEANet [13]	L.R	ESWA 24	33	3.27	9.62	0.8708	0.9207	0.9338	0.9417	0.8638	0.8624	0.8662	0.0312	0.8868	0.8373	0.7595
MSHNet [52]	L.R	CVPR 24	85	4.07	6.11	0.7659	0.8111	0.8402	0.8439	0.7075	0.7119	0.7199	0.0937	0.7046	0.8143	0.6298
ISAANet (Ours)	L.R	TGRS 24	108	3.59	4.34	**0.8732**	**0.9239**	**0.9431**	**0.9497**	**0.8694**	**0.8696**	**0.8760**	**0.0281**	**0.8873**	0.8511	**0.7724**

UIU-Net and MSHNet are designed to accurately capture tiny targets in infrared images, making full use of the unique information structure and features of such images. However, conventional remote sensing images contain targets of different sizes, which makes these methods fail to be effectively applied to the detection of large-size targets, resulting in their poor performance on the three datasets. This comparative experiment emphasizes the need for the development of adaptable and efficient models that take into account a wide range of target sizes in the field of remote sensing salient object detection.

Some models (e.g., PA-KRN and DNTDF) originally designed for natural scenes suffer from unsatisfactory evaluation scores after being transferred for remote sensing scenarios, which are underperforming the models (e.g., ACCoNet and AESINet) specifically designed and optimized for remote sensing data. This observation confirms that models trained on natural scenes lack robust generalization capabilities when

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deployed to the remote sensing domain. Hence, the development of models specialized for remote sensing scenes becomes imperative. Moreover, it is also worth noting that the models with a large number of parameters, such as ICON and GeleNet, perform well in terms of accuracy, but their running efficiency is relatively low. Conversely, the lightweight alternatives, represented by CorrNet and SeaNet, sacrifice some accuracy for higher speed. This signifies that both cumbersome and lightweight models inherently retain their respective advantages and limitations. In summary, our proposed model effectively achieves a favorable balance between high accuracy and efficiency to harness the complementary strengths of both types of models in the comprehensive performance comparison.

We observe that some models such as VST, ICON, and GeleNet adopt advanced transformer structures as their backbones, which turns outperformance over other models with CNN backbones. Surprisingly, our model can rival these transformer-based models. It achieves the lowest MAE scores of 0.0077–0.0063 on the ORSSD and EORSSD datasets, respectively, superior over GeleNet. Even on the large-scale ORSI-4199 dataset, the gap in MAE between it and GeleNet is only 0.0017, further narrowing the performance gap between the lightweight and the nonlightweight models. Compared with other models based on CNN backbones, ISAANet has higher inference quality across three datasets. For instance, ISAANet surpasses ACCoNet and AESINet by 1.37%–1.32% on average for the E_{ξ}^{adp} metric, respectively. Similarly, in terms of the F_{β}^{adp} score, ISAANet also has a leading edge of 3.21% and 4.93% over the two competitors, respectively. The statement on the comparison of ISSANet with other lightweight models can be found in Section I of the supplemental material.

Moreover, we compare model capacity and efficiency. Compared with GeleNet, our model runs 78 FPS faster and reduces parameters and FLOPs by 7× and 2.7×, respectively. Under these conditions, the performance gap between our model and GeleNet is marginal on three datasets, confirming that lightweight models can also achieve high accuracy detection through the iterative encoding–decoding paradigm. Besides, ISAANet only has approximately 1/29 of the parameters of ACCoNet, but it achieves better evaluation results, which also validates its effectiveness. Most remarkably, even compared with other lightweight models, our approach attains the fastest inference speed, further highlighting the superiority of our method. Therefore, by simultaneously improving both efficiency and reasoning quality, the proposed ISAANet exhibits higher applicability. In addition, GPU memory consumption for various methods can be found in Section II of the supplemental material.

2) Qualitative Comparison: For visual comparisons of inference quality, we present the saliency maps generated by ISAANet and some state-of-the-art methods in Fig. 2 of the supplemental material. These scenes are chosen for their representativeness, enabling the examination of the generalization performance of different models. In rows 1–4, where salient objects exhibit small-scale characteristics, ISAANet accurately segments them out. In contrast, other methods show limited perceptual abilities toward tiny objects, resulting in imprecise localization of targets in their output predictions. Moreover, some models exhibit confusion by background objects, leading to faulty reasoning and background noise interference in their saliency predictions. Detecting tiny objects has remained an inherently challenging task. Our approach of iteratively adjusting the encoder enables continuously enhancing the model's perception of specific targets. Empowered by this advantage, small objects can be effectively highlighted in our saliency maps. Notably, in rows 1 and 2, the structures of salient objects are quite delicate, making it challenging to maintain clear boundaries in the segmentation outputs. Compared with other methods, ISAANet better preserves the integrity of object boundaries. This boon is owed to the boundary-guided iterative updates imposed on the encoder using boundary prompt, steering its attention toward critical boundary signals.

In rows 5–8, scenes with salient objects of regular sizes are presented. The majority of the existing methods only provide rough identifications for portions of the salient objects, leading to incomplete objects detections. In multiobject scenarios (rows 4 and 5), some methods, such as MEANet, SAMNet, and CorrNet, struggle to detect certain objects with relatively less bright color appearances. In contrast, our approach consistently performs well in these examples and provides more complete segmentation of salient objects. This is primarily attributed to the HIDI module, effectively integrating multilevel features and preventing the oversight of informative cues.

The last four rows of Fig. 2 in the supplemental material depict scenes with large-sized objects of diverse shapes. In the saliency maps generated by the existing methods, the salient regions are neither fully nor uniformly segmented. As shown in the ninth row, most methods lack a comprehensive perception of the salient objects, resulting in their saliency maps that only preserve partial edge regions of the targets. When foreground objects share similar color appearances with the background (see the last row), ISAANet maintains a high perceptual awareness and outputs more accurate segmentation results. Conversely, reasoning by other models is far from ideal, and they fail to recognize large parts of the foregrounds. More severely, CorrNet and CSNet incorrectly identify the entire scene as background, and no salient regions exist in their inferences.

C. Ablation Studies

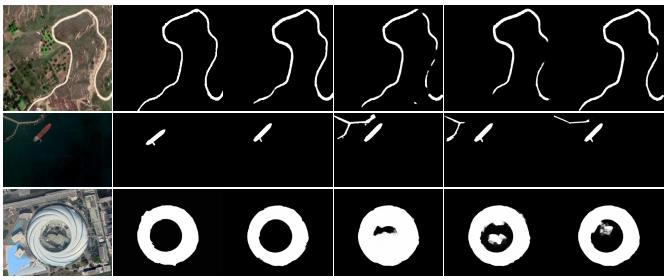
1) Effectiveness of ISAANet: To evaluate the effectiveness of the proposed iterative encoding–decoding architecture, we conducted ablation experiments by varying the number of inference iterations (Z) and examining the model performance at different Z values. When Z is set to 1, the architecture reverts to a standard encoder–decoder configuration without iterative optimization, which serves as the baseline. As Z increases to 2 and 3, the iterative process is introduced, where the encoder features can be refined under the guidance of preceding decoder feedback. As can be seen in Table III, superior performance over the baseline is obtained when Z is set to 3, with the F_{β}^{adp} metric being boosted by 2.47% and 1.50% on the ORSSD and ORSI-4199 datasets, respectively. This confirms the beneficial effect of circulating saliency information in facilitating reasoning

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TABLE III

PERFORMANCE COMPARISON AFTER SETTING DIFFERENT NUMBER (Z) OF ITERATIONS. THE BEST SCORES ARE HIGHLIGHTED IN **BOLD**

Settings	#Param (M)↓	FLOPs (G)↓	ORSSD [47]			ORSI-4199 [3]		
			$F_\beta^{\text{adp}}\uparrow$	$E_\xi^{\text{adp}}\uparrow$	$S_\alpha\uparrow$	$F_\beta^{\text{adp}}\uparrow$	$E_\xi^{\text{adp}}\uparrow$	$S_\alpha\uparrow$
$Z=1$	3.35	1.69	0.8735	0.9693	0.9269	0.8544	0.9168	0.8659
$Z=2$	3.47	3.02	0.8879	0.9745	0.9356	0.8623	0.9216	0.8710
$Z=3$	3.59	4.34	**0.8982**	**0.9777**	**0.9429**	0.8694	**0.9239**	0.8732
$Z=4$	3.71	5.66	0.8920	0.9756	0.9410	**0.8697**	0.9234	**0.8742**



(a) (b) (c) (d) (e) (f)

Fig. 9. Saliency maps output by ISAANet with varying numbers (Z) of iterations. (a) Input. (b) GT. (c) $Z=3$. (d) $Z=1$. (e) $Z=2$. (f) $Z=4$.

and comprehension. However, further increases in Z do not necessarily yield continued improvements. For instance, when Z equals 4, the increased complexity in network structure and parameters may lead to overfitting and even performance degradation, as observed by S_α score dropping from 0.9429 to 0.9410 on the ORSSD dataset. At the same time, the number of model parameters increases from 3.59 to 3.71 MB, accompanied by a rise in FLOPs from 4.34 to 5.66 G. This escalation in both parameters and computational operations results in an increased demand for computational resources and a concurrent decrease in operational efficiency. In addition, some saliency maps produced by networks consisting of different iterations are shown in Fig. 9. Among these inference results, the saliency maps output by the network corresponding to $Z=3$ are more accurate, with the fine-grained structures being well-preserved. Based on the tradeoff between performance and computational resource consumption, selecting the number of iterations as 3 is evidently an ideal compromise that ensures high detection performance while considering computational efficiency. It should be noted that the selection of the optimal number of iterations may be influenced by various factors such as the specificity of the dataset, the model architecture, and the specific requirements of the task. In practical applications, further adjustments and optimizations may be needed based on specific situations.

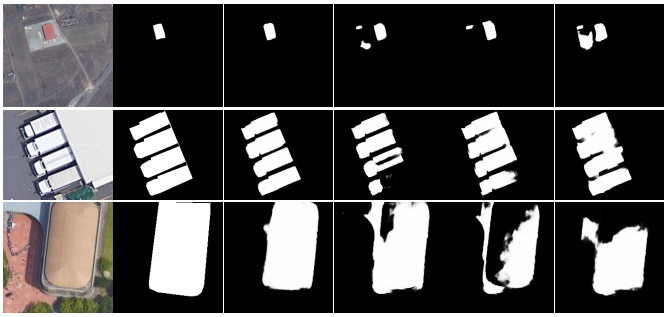
Besides, ablation experiments on setting the foreground segmentation threshold and the proposed saliency assignment (SA) strategy can be found in Section IV of the supplemental material.

2) Contribution of the HIDI Module: The presented HIDI module is validated through comparisons with various common feature fusion methods, including elementwise addition (ADD), multiplication (MUL), and simple channel concatenation (CAT). From Table IV, on the ORSSD dataset,

TABLE IV

EVALUATION COMPARISONS BETWEEN HIDI MODULE AND COMMON FUSION OPERATIONS. THE BEST SCORES ARE HIGHLIGHTED IN **BOLD**

Settings	ORSSD [47]			ORSI-4199 [3]		
	$F_\beta^{\text{adp}}\uparrow$	$E_\xi^{\text{adp}}\uparrow$	$S_\alpha\uparrow$	$F_\beta^{\text{adp}}\uparrow$	$E_\xi^{\text{adp}}\uparrow$	$S_\alpha\uparrow$
ADD	0.8843	0.9714	0.9326	0.8592	0.9182	0.8686
MUL	0.8865	0.9731	0.9341	0.8601	0.9188	0.8690
CAT	0.8857	0.9723	0.9335	0.8585	0.9173	0.8679
HIDI	**0.8982**	**0.9777**	**0.9429**	**0.8694**	**0.9239**	**0.8732**



(a) (b) (c) (d) (e) (f)

Fig. 10. Saliency maps produced using different fusion operations in the decoder. (a) Input. (b) GT. (c) HIDI. (d) ADD. (e) CAT. (f) MUL.

E_ξ^{adp} scores for elementwise addition, multiplication, and simple concatenation are 0.9714, 0.9731, and 0.9723, respectively. Notably, the HIDI module achieves a higher E_ξ^{adp} score of 0.9777, demonstrating improvements of 0.63%, 0.46%, and 0.54%, respectively. Similar improvements are observed on the ORSI-4199 dataset, underscoring the superiority of the HIDI module. This advantage arises from the hierarchical information fusion strategy, which dynamically emphasizes useful combinations while suppressing the influence of ineffective ones. In contrast, traditional operations lack this discriminatory capability, treating all information as equally important and leading to potential redundant interference and an inability to accurately capture effective components across feature layers. In Fig. 10, the visual comparisons between the HIDI module and three alternative fusion methods are presented. After incorporating the HIDI module, the network generates saliency maps with a better effect in localizing salient objects and efficiently reduces interference from extraneous noise. Overall, salient regions are more clearly and uniformly segmented in the results processed by the HIDI module.

We also examine the influence of varying the number of feature splits (N) on model performance, as shown in Table V. As N increases from 2 to 4, we do not observe consistent performance enhancements. On the contrary, we see that when N is increased to 3, the performance scores of the evaluation metrics begin to deteriorate consistently across both the datasets. For example, F_β^{adp} and E_ξ^{adp} scores exhibit an average decline of 0.59% and 0.28%, respectively. Excessive feature partitioning may pose challenges for weight regression learning in two aspects. First, reduced semantic richness

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TABLE V
INFLUENCE OF THE NUMBER (N) OF FEATURE SPLITS IN THE HIDI MODULE. THE BEST SCORES ARE HIGHLIGHTED IN **BOLD**

Settings	ORSSD [47]			ORSI-4199 [3]		
	$F_\beta^{\text{adp}} \uparrow$	$E_\xi^{\text{adp}} \uparrow$	$S_\alpha \uparrow$	$F_\beta^{\text{adp}} \uparrow$	$E_\xi^{\text{adp}} \uparrow$	$S_\alpha \uparrow$
$N = 2$	**0.8982**	**0.9777**	**0.9429**	**0.8694**	**0.9239**	**0.8732**
$N = 3$	0.8930	0.9740	0.9413	0.8629	0.9220	0.8713
$N = 4$	0.8894	0.9726	0.9398	0.8613	0.9206	0.8701

within each subunit weakens supervisory signals needed for weight learning. Second, because the model needs to learn the regression of more weights, it is prone to converging on suboptimal solutions during network training, especially with limited training samples. The joint impact of these factors can hinder the effect of weight learning and assignment. Therefore, selecting an appropriate N value is crucial to ensure each subunit contains sufficient semantics to support relationship learning while avoiding an excessive introduction of learnable parameters. Under the guidance of the above analysis, we identify $N = 2$ as the optimal setting for the HIDI module.

Furthermore, results on the comparison of the HIDI module with other fusion modules and the analysis of dynamic weight assignment can be found in Section V of the supplemental material.

D. Other Experiments

Analysis of failure cases, robustness evaluation of ISAANet, and generalizability and application discussion can be found in Sections VI–VIII of the supplemental material, respectively.

V. CONCLUSION

To address the predicament of prevailing lightweight models that achieve high efficiency at the cost of low accuracy, we propose an ISAANet for efficient salient object detection in remote sensing images. The key innovation of ISAANet lies in establishing a bidirectional information flow mechanism between the encoder and the decoder, breaking the limitations of the traditional unidirectional inference structure. By iteratively updating features based on feedback from the decoder's inference results, the encoder provides more accurate saliency cues, which drives the decoder to generate higher quality detection results. Furthermore, we devise a novel HIDI module for cross-layer feature fusion. The HIDI module uses a grouping strategy and a quality evaluation mechanism to assign adaptive weights to different feature groups, which can effectively suppress redundant interference while focusing on crucial complementary information. Extensive experiments show that the ISAANet model achieves a favorable balance between accuracy and efficiency with fewer parameters. On one hand, it narrows the reasoning quality gap with the cumbersome models. On the other hand, ISAANet runs at a higher speed of 108 FPS.

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